

MindEye: A VR Imagery Training System for Individuals with Acquired Visual Imagery Impairments and Mild Aphantasia

Tian Wang*
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
tian57730@gmail.com

Chenyang He*
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
hechenyang1111@163.com

Ruici Song
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
ruicisong@gmail.com

Jinghan Zhang
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
jhzhzhang008823@gmail.com

Yijing Xu
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
1325281823@qq.com

Pei Zhang[†]
Shenzhen Polytechnic University
Shenzhen, China
ppppzhangpp@hotmail.com

Mengyao Guo
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
guomengyao@hit.edu.cn

Guanyou Li
Harbin Institute of Technology,
Shenzhen
Shenzhen, China
804064958@qq.com

Zhiwei Deng
Configreality Technology Co.Ltd.
Shenzhen, China
linkerwei@configreality.com

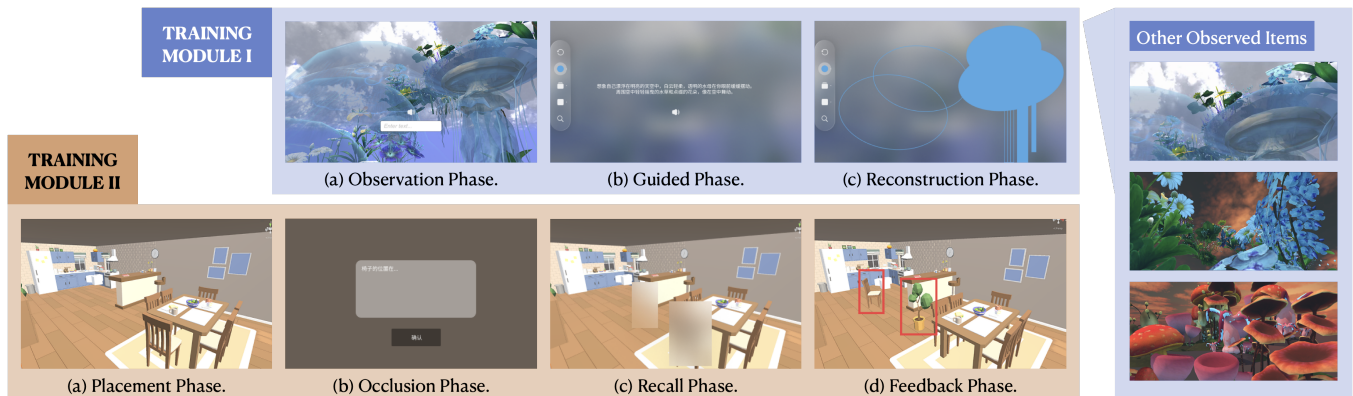


Figure 1: Two training modules in *MindEye*.

Abstract

Visual imagery deficits, including aphantasia and acquired imagery impairments, significantly impact cognitive functioning yet lack effective interventions. We present *MindEye*, a VR-based training system combining guided imagery reconstruction with spatial-semantic association techniques to rebuild visual imagery pathways. The system features two modules that provide structured

verbal cues and progressive visual scaffolding, and that preserve spatial cognition to strengthen imagery-memory connections. Our study with 6 participants demonstrated significant improvements across all measures: imagery clarity increased from 2.3 to 3.7, scene reconstruction from 52% to 76%, and spatial recall from 58% to 85%, with participants having initially low imaging ability showing the largest gains. These results suggest that combining guided imagery training with spatial-semantic association offers a promising therapeutic approach for individuals with imagery deficits.

*Both authors contributed equally to this research.

[†]Pei Zhang is the corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771463>

ACM Reference Format:

Tian Wang, Chenyang He, Ruici Song, Jinghan Zhang, Yijing Xu, Pei Zhang, Mengyao Guo, Guanyou Li, and Zhiwei Deng. 2025. *MindEye: A VR Imagery Training System for Individuals with Acquired Visual Imagery Impairments and Mild Aphantasia*. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771463>

1 Introduction

Visual imagery ability, the capacity to generate and manipulate mental images without external input, plays a crucial role in memory, planning, and emotional regulation [Liu et al. 2025]. However, this fundamental capacity is severely compromised in individuals with aphantasia, characterized by reduced or absent visual imagery that impairs autobiographical memory, emotional regulation, learning efficiency, and empathy [Wicken et al. 2021], creating significant daily challenges. Virtual Reality (VR) offers unprecedented potential for addressing aphantasia through controlled, immersive experiences that can simulate realistic perceptual experiences and activate brain regions associated with imagery (e.g., hippocampus, prefrontal cortex) [Moll and Sykes 2023], suggesting VR-simulated experiences can reactivate dormant imagery pathways. Moreover, recent advances in generative AI enable real-time scene creation and adaptive content generation, while VR's programmable nature enables systematic training protocols with precise feedback. We introduce *MindEye*, a VR-based imagery training system for individuals with acquired visual imagery impairments and mild aphantasia.

2 MindEye

MindEye uses immersive virtual reality and AI-generated adaptive content to restore visual imagery abilities in individuals with acquired imagery impairments and mild aphantasia. It deploys the perceptual-imagery equivalence principle through immersive perceptual activation and structured tasks to restore image generation and manipulation abilities via two training modules: **Training Module I - Guided Imagery Module**: Combines visual presentation with verbal guidance to stimulate visual cortices and enhance imagery construction; **Training Module II - Spatial-Object Path Training Module**: Leverages preserved spatial navigation through a cycle of "placement-occlusion-recall-feedback" to integrate spatial, semantic, and visual information.

2.1 Modules Explanation

Training Module I activates visual imagery pathways through verbal guidance and immersive multisensory experiences. Since visual imagery and perception share neural substrates in the primary visual cortex and higher-order areas [Pearson et al. 2008], simulated perception can activate internal imagery, while combined verbal-visual presentation enhances mental image formation. This module implements three sequential phases. The Observation Phase presents immersive VR scenes with simultaneous voice guidance to stimulate cognitive associations between visual and auditory information (Figure 1 Module Ia). During the Guided Phase, visual content gradually fades while verbal descriptions continue, requiring users to mentally reconstruct the scene using linguistic cues and developing imagery skills (Figure 1 Module Ib). The Reconstruction Phase challenges users to recreate the scene in an empty virtual space using VR controllers to select and position visual elements (Figure 1 Module Ic).

Training Module II implements a "placement-occlusion-recall-feedback" cycle based on spatial cognition theory and the Method of Loci. Since spatial processing abilities are often preserved in

individuals with imagery impairments, this system leverages intact spatial encoding to construct "space-semantic-imagery" linkage pathways that compensate for deficits in visual representation [Dresler et al. 2017; Maguire et al. 2000; Zeman et al. 2010]. Users explore familiar VR environments and place representative objects at specific locations following system prompts. The system then occludes objects and asks users to recall both content and location before re-presenting objects for feedback. Training gradually establishes space-semantic-context associations while recording navigation and recall performance. In later stages, users recall all placed objects in sequence, forming a Mental Route Map that activates hippocampal-prefrontal-parietal neural pathways (Figure 1 Module II).

2.2 Technical Implementation

The system was developed in Unity 2023.1.16f1 and deployed on Meta Quest 3. The XR Interaction Toolkit enables hand controllers to grab objects, perform ray-based selection, click UI elements, and receive visual feedback, supporting object placement in Module I and semantic positioning in Module II. Spatial navigation is handled by NavMesh, while Timeline and Cinemachine manage scene transitions and guided tours. TextMeshPro displays task prompts and feedback, and Addressables enable on-demand loading of scenes and objects. User actions are recorded via JSONUtility, and voice guidance and recognition are provided through Baidu TTS and ASR. The DeepSeek API dynamically generates scene code to support adaptive training based on user input and imagery performance.

3 User Study

To evaluate the effectiveness of our system in enhancing visual imagery in individuals with acquired imagery deficits or mild aphantasia, we conducted a user study involving 6 participants aged 18-40 ($M=28.5$), including PTSD patients, self-reported imagery-impaired users, and healthy controls. Participants demonstrated measurable improvements following training, with self-reported imagery clarity increasing from 2.3 to 3.7 on a 1-5 Likert scale, scene reconstruction similarity scores improving from 52% to 76% after three sessions, and object-location recall accuracy rising from 58% to 85%. Participants with initially low imaging ability showed the largest gains, with feedback revealing that verbal guidance was most effective for initiating visual recall and spatial navigation, enhancing memory organization.

4 Conclusion and Futurework

In this paper, we present a VR-based training system combining guided imagery and spatial-semantic association techniques to address visual imagery deficits in individuals with aphantasia and acquired imagery impairments. Our user study showed that our dual-module approach effectively leverages preserved spatial cognition abilities while systematically rebuilding visual imagery pathways through structured training cycles. As for the future development, we will validate the system's feasibility through controlled studies with standardized assessment tools (e.g., VVIQ), enhance user experience and training effectiveness for personalized training, and recruit more participants with imagery impairments for long-term programs.

References

- Martin Dresler, William R Shirer, Boris N Konrad, Nils CJ Müller, Isabella C Wagner, Guillén Fernández, Michael Czisch, and Michael D Greicius. 2017. Mnemonic training reshapes brain networks to support superior memory. *Neuron* 93, 5 (2017), 1227–1235.
- Jianghao Liu, Minye Zhan, Dounia Hajhate, Alfredo Spagna, Stanislas Dehaene, Laurent Cohen, and Paolo Bartolomeo. 2025. Visual mental imagery in typical imagers and in aphantasia: A millimeter-scale 7-T fMRI study. *cortex* 185 (2025), 113–132.
- Eleanor A Maguire, David G Gadian, Ingrid S Johnsrude, Catriona D Good, John Ashburner, Richard SJ Frackowiak, and Christopher D Frith. 2000. Navigation-related structural change in the hippocampi of taxi drivers. *Proceedings of the National Academy of Sciences* 97, 8 (2000), 4398–4403.
- Brigham Moll and Ed Sykes. 2023. Optimized virtual reality-based Method of Loci memorization techniques through increased immersion and effective memory palace designs: A feasibility study. *Virtual Reality* 27, 2 (2023), 941–966.
- Joel Pearson, Colin WG Clifford, and Frank Tong. 2008. The functional impact of mental imagery on conscious perception. *Current Biology* 18, 13 (2008), 982–986.
- Marcus Wicken, Rebecca Keogh, and Joel Pearson. 2021. The critical role of mental imagery in human emotion: Insights from fear-based imagery and aphantasia. *Proceedings of the royal society B* 288, 1946 (2021), 20210267.
- Adam ZJ Zeman, Sergio Della Sala, Lorna A Torrens, Viktoria-Eleni Gountouna, David J McGonigle, and Robert H Logie. 2010. Loss of imagery phenomenology with intact visuo-spatial task performance: A case of 'blind imagination'. *Neuropsychologia* 48, 1 (2010), 145–155.